

Terp Protect

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Overview

This project, developed by Terp Protect, focuses on designing a comprehensive crime and arrest log database system for the University of Maryland Police Department (UMPD). The goal is to streamline how police records are stored, managed, and analyzed. By centralizing data related to arrests, cases, officers, and departments, the system enhances transparency, ensures data consistency, and enables efficient access to critical information. Through this database, UMPD can easily track offender demographics, monitor departmental performance, and analyze crime patterns across locations and time periods ultimately supporting data-driven decision-making and improving campus safety management.

Data Sources

Data Source: Arrest Logs provided by UMPD.

- **Data Processing Note:** For the purpose of this database project, we generated synthetic data based on the structure and format of actual UMPD arrest logs.
- **Reference URL:** <https://umpd.umd.edu/statistics-reports/arrest-logs>
- **Content Included:**
 - Arrest Number
 - Arrested Date and Time
 - UMPD Case Number
 - Arrestee demographics (Age, Race, Sex)
 - Arrested Charges (specific violations and statutes)
 - Annual arrest data organized by year
 - Arrestee information
 - Officer details
 - Department assignments
 - Case records
 - Charge categories
 - Arrest logs linking all entities

Database Schema & Table Design

The database design consists of the following tables:

Table Name	Key Fields	Description
Arrestee	arresteeId (PK), arresteeFirstName, arresteeLastName, arresteeDOB, arresteeSex, arresteeRace	Stores demographic information about individuals who have been arrested
Officer	officerId (PK), officerFirstName, officerLastName, departmentId (FK), officerLeadId (FK)	Contains officer information and departmental assignments
Department	departmentId (PK), departmentName, departmentType, departmentLocation, departmentPhoneNumber, officerHeadId (FK), startDate	Organizes officers by department with head officer tracking
Case	caseId (PK), caseType, caseReportedOn, caseClosedOn, caseDescription	Records details about reported cases
CaseChargeCategory	caseId (PK, FK), caseChargeCategory (PK)	Associates cases with one or more charge categories
ArrestLog	arrestLogId (PK), caseId (FK), arresteeId (FK), officerId (FK), arrestLogTimeStamp, arrestLocation	Central table linking arrests to cases, arrestees, and officers

Setup and Reference Files

File: [Project_501_04.sql](#)

SQL Scripts

Create Script:

- Creates all the tables we need for the database
- Sets up primary keys and foreign keys
- Includes all the referential integrity stuff (CASCADE, NO ACTION)

Insert Script

- Adds sample data to all our tables
- We made up synthetic data based on what real UMPD arrest logs look like
- Makes sure all the foreign keys match up correctly

Drop Script

- Deletes all the tables so we can start fresh
- Helpful when we need to reset everything and try again

How to Test the Project

What We Used

- Server: Microsoft SQL Server on [doitsqlx.rhsmith.umd.edu](#)
- Database: BUDT702_Project_0501_04

Steps to Run Everything

1. Drop the Tables

- First, run [Project_501_04.sql](#) to clear out any old tables
- This makes sure we're starting clean

2. Create the Tables

- Next, run the file to build all our tables
- This sets up the whole database structure with all relationships

3. Add the Data

- Then run the file to fill the tables with data
- Now the database has all the sample information we need for testing

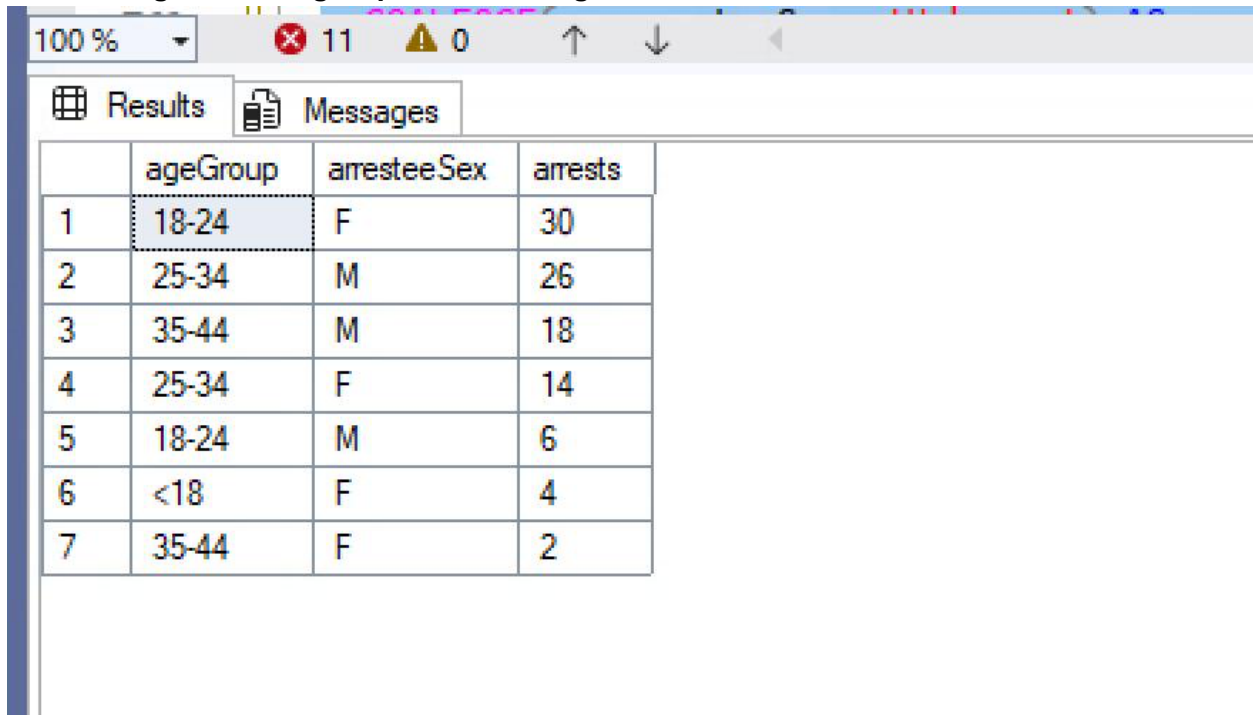
4. Test It Out

- Run some sample queries to make sure everything works
- Check that the data looks right and relationships are working

What We Tested

Mission Objectives Tests:

1. Which age and sex groups have the highest number of arrests?



	ageGroup	arresteeSex	arrests
1	18-24	F	30
2	25-34	M	26
3	35-44	M	18
4	25-34	F	14
5	18-24	M	6
6	<18	F	4
7	35-44	F	2

2. Determine the most common charge categories leading to arrests and how they vary over time.



Results



Messages

	arrestMonth	caseChargeCategory	arrestsWithCategory
51	2022-02-01	Phishing	1
52	2022-02-01	Possession	1
53	2022-02-01	Property Damage	1
54	2022-02-01	Public Disturbance	1
55	2022-02-01	Public Intoxication	1
56	2022-02-01	Reckless Driving	1
57	2022-02-01	Reckless Endangerm...	1
58	2022-02-01	Resisting Detention	1
59	2022-02-01	Restraining Order Viol...	1
60	2022-02-01	Robbery	1
61	2022-02-01	Tampering with Evide...	1
62	2022-02-01	Traffic Accident	1
63	2022-02-01	Unauthorized Access	1
64	2022-02-01	Unlawful Assembly	1
65	2022-02-01	Vandalism	1
66	2022-03-01	Fraud	2
67	2022-03-01	Animal Cruelty	1
68	2022-03-01	Arson	1
69	2022-03-01	Carving Without Permit	1

3. Calculate the number and proportion of reported cases that did not result in any arrest.

100 % No Issues Found

Results Messages

	casesWithNoArrest	totalCases	proportionNoArrest
1	5	105	0.05

4. Compute and compare the average case duration across departments to identify performance differences.

100 % No Issues Found

Results Messages

	departmentId	departmentName	avgCaseDurationDays	casesCounted
1	D108	Counter-Terror.	4.36363636363636	11
2	D103	Financial Crimes	6.27272727272727	11
3	D109	Organized Crime	7.44444444444444	9
4	D105	Forensic Science	9.16666666666667	12
5	D102	Homicide Invest.	10.9166666666667	12
6	D106	Traffic Control	11.25	12
7	D110	Comm. Outreach	11.4	10
8	D107	Special Victims	11.5454545454545	11
9	D104	Narcotics Div.	12.9166666666667	12

5. Identify which departments and department heads have recorded the highest number of arrests and handled the most cases.

	departmentId	departmentName	officerHeadId	departmentHead	totalArrests	distinctCasesHandled
1	D105	Forensic Science	OF0000004	Barbara Clark	12	12
2	D102	Homicide Invest.	OF0000001	Robert King	12	12
3	D104	Narcotics Div.	OF0000003	William Adams	12	12
4	D106	Traffic Control	OF0000005	Richard Lewis	12	12
5	D108	Counter-Terror.	OF0000007	Joseph Allen	11	11
6	D103	Financial Crimes	OF0000002	Linda Scott	11	11
7	D107	Special Victims	OF0000006	Susan Hall	11	11
8	D110	Comm. Outreach	OF0000009	Charles Hernandez	10	10
9	D109	Organized Crime	OF0000008	Patricia Young	9	9
10	D101	Cyber Crimes	OF0000040	Rachel Richardson	0	0

6. Determine the most frequently handled case types by each department.

100 %		NO ISSUES FOUND		
Results		Messages		
	departmentId	departmentName	caseType	handledCount
1	D110	Comm. Outreach	Fraud	2
2	D108	Counter-Terror.	Assault	2
3	D103	Financial Crimes	Shoplifting	2
4	D105	Forensic Science	Cybercrime	2
5	D102	Homicide Invest.	Burglary	2
6	D104	Narcotics Div.	Animal Cruelty	2
7	D109	Organized Crime	Illegal Firearms	2
8	D107	Special Victims	Theft	2
9	D106	Traffic Control	Disorderly Conduct	2

User Interface Development

After completing the database design and testing all our queries, we created a user-friendly web interface to visualize and interact with our arrest data system.

Front-End Dashboard

We generated a complete UI/UX design for the TerpProtect information system using Claude AI. The dashboard was designed to directly address our project mission and objectives by providing:

Key Features That Support Our Mission:

- **Interactive Dashboard:** Displays real-time statistics and visualizations that answer our key research questions
- **Data Visualizations:** Charts and graphs that show arrest trends, demographics, and charge categories - directly implementing our mission objectives
- **Search and Filter Tools:** Allow users to quickly find specific arrests, cases, or officers
- **Recent Arrests Table:** Shows the latest arrest logs with all relevant information
- **Quick Action Buttons:** Easy access to generate reports and analyze department performance

How the UI Connects to Our Objectives:

1. **Age/Gender Demographics (Objective 1):** The pie chart on the dashboard shows arrest distribution by age groups
2. **Monthly Charge Trends (Objective 2):** The bar chart displays most common charge categories
3. **Arrest Trends (Objective 1-6):** The line graph visualizes arrest patterns over time
4. **Department Performance (Objective 5):** Quick action cards link to officer and department analytics
5. **Case Analysis (Objective 4):** Direct access to case duration and resolution metrics

Design Philosophy

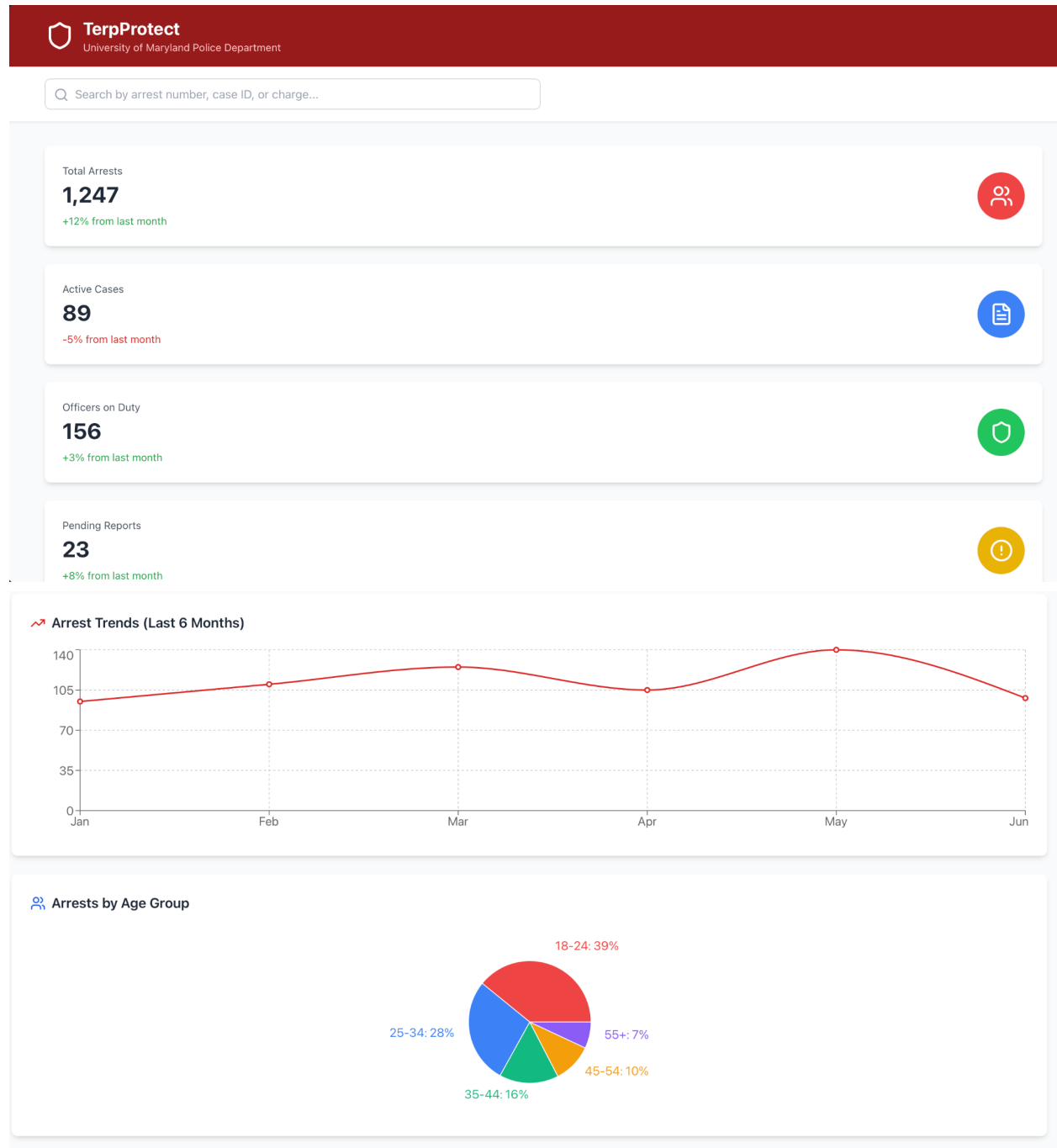
We designed the interface with the following principles in mind:

- **User-Friendly:** Easy navigation and clear information hierarchy
- **Professional:** Clean design appropriate for a law enforcement database
- **Informative:** All key metrics visible at a glance on the home page
- **Interactive:** Users can search, filter, and drill down into specific data
- **Responsive:** Works on desktop, tablet, and mobile devices

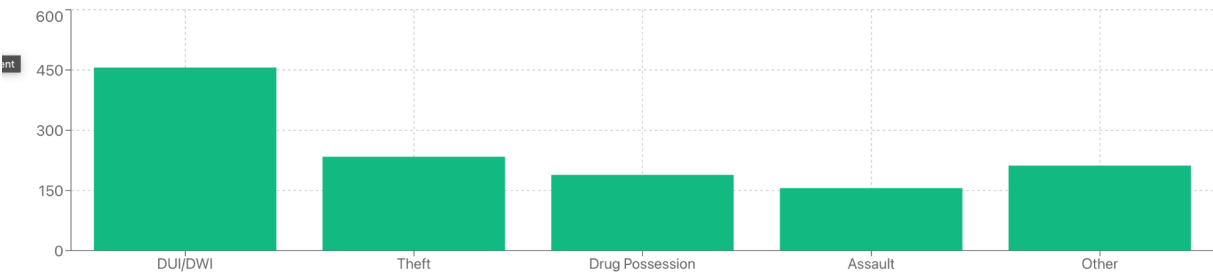
The UI successfully brings together all components of our database project - from the relational schema to the complex SQL queries - into an accessible interface that satisfies the needs of our project mission and objectives. Users can now easily explore arrest patterns, analyze department performance, and generate insights without needing to write SQL queries themselves.

Link to the user page: <https://claude.ai/public/artifacts/1823ca38-ef36-4625-be7f-639db562ea26?fullscreen=true>

Screenshots of the user interface of the Terp Protect:



📄 Most Common Charge Categories



📅 Recent Arrests

[View All >](#)

Arrest ID	Date	Name	Charge	Status	Action
24369	2025-01-07	John D.	Motor Vehicle/Unlawful Taking	Processed	View Details
24368	2025-01-07	Mike S.	Theft: \$100,000+	Pending	View Details
24355	2025-01-03	David R.	DUI - Alcohol & Drugs	Processed	View Details
24346	2025-01-08	Robert M.	Theft Scheme	Under Review	View Details

Generate Report

Create custom analytics reports



Officer Performance

View department statistics



Case Analysis

Analyze case resolution times



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All data is synthetic and for educational purposes only

References:

Data Sources

- University of Maryland Police Department, Arrest Logs, available at <https://umpd.umd.edu/statistics-reports>

Tools & Assistance

- **ChatGPT (OpenAI)**
 - Understanding the structure and format of UMPD arrest log data
 - Generating synthetic data that resembles real arrest records while protecting privacy
 - Creating **INSERT** statements to populate database tables
 - Reviewing database design and suggesting improvements
- **Claude AI**
 - Formatting and organizing project documentation
 - User Interface Development
- **Lucidchart**
 - Creating the Entity-Relationship (ER) diagram
- **Microsoft Word**
 - Drafting and finalizing the database design document
- **Canva**
 - Designing the project presentation slides

Note on Synthetic Data

All data in this database is completely synthetic and was generated using AI tools. We created this data to match the structure of real UMPD arrest logs but with fictional names, dates, and case details. No actual arrest records or personal information were used in this project.